## AI Model Openness - Reducing Vendor Lock-in

*Condensed comparison of open-weight models to reduce AI vendor lock-in*

### Executive Summary

- Relying on one proprietary AI provider creates vendor lock-in - it raises long-term costs and limits control, flexibility, and negotiating power.
- This report explains what "openness" means for AI models (open weights, code, data, training methods, evaluation, and licensing) and compares 10 leading open-weight models against that framework.
- Open-weight models - especially **Qwen 3.5, DeepSeek V4, GLM-5.2, and Mistral Large 3** - are now close to closed frontier models (Claude, GPT, Gemini) on coding and agentic benchmarks, at a fraction of the API cost.
- Self-hosting removes vendor lock-in but adds a real cost: GPU infrastructure, ops staffing, and 24/7 maintenance. It only becomes cheaper than an API at high, sustained usage (tens of billions of tokens/months).
- **Recommendation:** Adopt **GLM-5.2** as the primary model for coding, software development, and technical workflows due to its strong engineering capabilities, cost efficiency, and deployment flexibility. Retain access to proprietary models for specialized reasoning tasks, while using self-hosted open-weight models for most development and automation workloads.

### 2. Comparing Leading Open-Weight Models

#### 2.1 Shortlist and Level of Openness

Each model below is checked against the weights available, is the code open, can it be self-hosted, and is commercial use allowed (and under what conditions).

| Model | Weights | License | OSS / Serving | Open Data | Methodology | Evaluation | Overall Openness |
|---|---|---|---|---|---|---|---|
| **Llama 4 Maverick** | 17B/400B | **Custom** — 700M-MAU cap, EU vision limit | vLLM, Ollama, TGI, NIM | Not released (~22T tok, undisclosed mix) | Model card only | Vendor + 3rd-party (lower) | Low |
| **Qwen 3.5** | 17B/397B | **Apache 2.0**, no caps | vLLM/SGLang | Not released | Blog + survey, no paper | Vendor-reported | Med |
| **Mistral Large 3** | 41B/675B | **Apache 2.0** | vLLM, NVFP4 | Not shared | Release post only | Vendor-reported | Low-Med |
| **Gemma 4** | 3.8–4B/26–31B | **Apache 2.0**; name/trademark restricted | vLLM, Ollama, RTX | Described, not released | Full tech report | Human-rater + benchmarks | Med-High |
| **DeepSeek V4 Pro/Flash** | 13–49B/284B–1.6T | **MIT**, unrestricted | vLLM/SGLang + eval sandbox | Not released; pipeline documented | Detailed tech report | Vendor + independent | High |
| **GLM-5.2** | 40B/~744B | **MIT**, no regional limits | vLLM/SGLang | Not disclosed | Partial write-ups | Largely self-reported | Med |

| Model | Weights | License | OSS / Serving | Open Data | Methodology | Evaluation | Overall Openness |
|---|---|---|---|---|---|---|---|
| **Kimi K2.6** | 32B/1T | **Modified MIT** >$20M/mo limit | vLLM, SGLang, KTransformers | Not released | Partial (MuonClip) | Vendor-controlled | Low-Med |
| **MiniMax M3** | ~23B/428B | **Custom** — authorization needed >$20M/yr | SGLang | Undisclosed | Full tech report | Vendor-run (flagged) | Low |
| **Nemotron 3 Ultra** | 55B/550B | **NVIDIA Open Model License** (not OSI) | vLLM, SGLang, Ollama, llama.cpp | Released (multiple datasets) | Full report + training code | Reproducible (NeMo SDK) | Highest |

A few things worth flagging from this breakdown:

- **Data transparency is the industry's weak point - Nemotron 3 Ultra** is the only model here that actually ships its pretraining datasets; everyone else describes the data mix in prose without releasing it.
- **"Vendor-run benchmarks" is the norm, not the exception -** Llama 4, GLM-5.2, and MiniMax M3 all have explicit third-party callouts noting their headline scores aren't independently verified.
- **License nuance is where the real differentiation is —** Qwen 3.5, Mistral Large 3, DeepSeek V4, and GLM-5.2 are the only truly unrestricted picks (Apache 2.0 or plain MIT); Kimi K2.6's revenue-based clause and MiniMax M3's authorization requirement are easy to miss if you only check "open weight = yes."

### 2.2 Benchmark Comparison

Figures are the best publicly available numbers as of this report's last-verified date; some are vendor-reported rather than independently confirmed (marked *). Treat the table as directional and re-verify before purchasing decisions.

| Model | SWE-Bench | BenchLM rank | BenchAlign score (/100) | GPQA Diamond | MMLU-Pro |
|---|---|---|---|---|---|
| Llama 4 Maverick | 53.0 | #205 / 215 | 22.7 | n/a | ~80–85 |
| Qwen 3.5 | 76.4 | #79 / 215 | 56.08 | n/a | n/a* |
| Mistral Large 3 | n/a | — | 49.57 | ~44 | 73.1 |
| DeepSeek V4 Pro | 80.6 | #48 / 215 | 60.0 | 90.1 | 87.5 |
| GLM-5.2 | 62.1 | #40 / 215 | 62.97 | 91.2 | n/a* |
| Kimi K2.6 | 80.2 | #65 / 215 | 58.73 | n/a | n/a* |
| MiniMax M3 | 80.5 | #18 / 215 | 68.8 | n/a | n/a* |
| Nemotron 3 Ultra | 71.9 | #157 / 215 | 43.67 | 87 | n/a* |

## 3. Cost — API vs. Self-Hosting, by Model

| Model | API cost ($/M in–out) | Self-hosting cost/mo (24/7, GPU rental only) | Self-hosting requirements | Where it wins |
|---|---|---|---|---|
| Gemma 4 (26–31B) | Free to self-host, no first-party API | ~$950–1,800 | 1 GPU | Cheapest to self-host |
| Llama 4 Maverick | $0.15–0.50 / $0.15–0.50 | ~$3,000–9,000 | 2-4 GPUs | Multimodal + broad tooling support |
| Qwen 3.5 | ~$0.11 / $0.11 | ~$11,500–23,000 | 8×H100 | Best cost-to-performance overall |
| DeepSeek V4 Flash | ~$0.14 / $0.28 | ~$11,500–23,000 | 8×H100 | Cheap API, strong reasoning |
| DeepSeek V4 Pro | ~$0.44 / $0.87 | $20,000+ | Multi-node | Highest open-weight benchmark scores |
| Mistral Large 3 | $0.50 / $1.50 | ~$11,500–23,000+ | Multi-node | Strong enterprise reasoning |
| GLM-5.2 | ~$1.40 / $4.40 | ~$11,500–23,000+ | 8× H200 | Strong coding/agentic benchmarks |
| Kimi K2.6 | ~$0.60–0.95 / $3.00–4.00 | ~$11,500–23,000+ | 8× H200 | Flexible multi-provider hosting |
| MiniMax M3 | $0.30–0.60 / $1.20–2.40 | $20,000+ | Datacenter-class | Long-context, high BenchAlign rank |
| Nemotron 3 Ultra | $0.50 / $2.20 | $20,000+ | 8×H200 or 16×H100 | NVIDIA-native deployments |
| Closed frontier (Claude Opus 4.8 / GPT-5.5) | ~$6–10 / $30–50 (per 500M-token estimate: $3,000–5,000/mo) | n/a - API only | n/a - API only | Hardest reasoning tasks, zero ops |

### 3.1 What Our Current Hardware Can Run

Our dev machine (Lenovo Legion 5, Intel i7-14650HX, 64GB RAM, RTX 4060 Laptop 8GB VRAM) is solid for local testing but is consumer laptop hardware, not a server. Its 8GB of GPU VRAM is the ceiling for anything run entirely on-GPU.

- ∉ None of the frontier open-weight models (Llama 4, Qwen 3.5, Mistral Large 3, DeepSeek V4, GLM-5.2, Kimi K2.6, MiniMax M3) fit on this machine, even at 4-bit quantization - all need multi-hundred-gigabyte, multi-GPU setups.
- ∉ CPU-offloaded inference (spilling layers into system RAM) is possible but slow - expect low single-digit tokens/second past ~13B parameters, not usable for production.
- ∉ What this machine is good for: local development and testing small models (Gemma 4 E2B/E4B and similar 2–4B models fit comfortably at int4).

## 4. Recommendation

### By capability (best for coding/technical workloads)

**GLM-5.2**: highest coding score among open-weight models on BenchLM, MIT license, flexible deployment, attractive cost-to-performance. **Qwen 3.5** is a strong secondary for general-purpose workloads (Apache 2.0, best overall value).

### By openness (best for enterprise audit & compliance)

- **Primary: Nemotron 3 Ultra.** Only model with released training datasets, a full technical report, published training code, and reproducible evals - the most auditable stack end-to-end. License isn't OSI-approved, so route through legal.
- **Secondary: DeepSeek V4 Pro/Flash.** Plain MIT, no usage caps, unusually detailed technical disclosure. Chinese-lab origin - confirm data-residency/compliance fit first.

### Bottom line

*Capability and openness point to different models, not a conflict, just different questions.*

- *Use GLM-5.2 as the default for coding and technical workloads*
- *Use Nemotron 3 Ultra where auditability and compliance defensibility matter most.*

*Reserve closed frontier APIs for the hardest reasoning tasks or where zero ops overhead is required.*

### References

*Key sources for the claims, benchmarks, and licensing details in this report. Re-verify before relying on any figure for a purchasing or compliance decision - several prices and models changed within the last 8 weeks.*


[1] Llama 4 license terms (700M MAU clause, EU restriction) — https://wcr.legal/llama-3-license-700m-mau-limit/

[2] Meta — The Llama 4 herd (official announcement) — https://ai.meta.com/blog/llama-4-multimodal-intelligence/

[3] Gemma Terms of Use (official) — https://ai.google.dev/gemma/terms

[4] Qwen 3.5 architecture, license, benchmarks — https://www.morphllm.com/qwen-3-5

[5] Mistral 3 / Large 3 official announcement — https://mistral.ai/news/mistral-3/

[6] DeepSeek-V4 technical report (arXiv) — https://arxiv.org/pdf/2606.19348

[7] DeepSeek V4 architecture, benchmarks, pricing — https://www.morphllm.com/deepseek-v4

[8] GLM-5.2 architecture, benchmarks, pricing — https://www.morphllm.com/glm-5-2

[9] Kimi K2.6 pricing and deployment guide — https://deepinfra.com/blog/kimi-k2-6-pricing-guide-deployment-tradeoffs

[10] MiniMax M3 architecture, benchmarks, pricing — https://www.morphllm.com/minimax-m3

[11] Nemotron 3 Ultra benchmarks, pricing & speed — https://benchlm.ai/models/nemotron-3-ultra

[12] H100/A100 cloud GPU pricing comparison (2026) — https://www.cloudzero.com/blog/h100-gpu-cost/

[13] Claude API pricing (Anthropic, 2026) — https://www.finout.io/blog/anthropic-api-pricing

[14] BenchLM.ai — model BenchAlign scores and rankings — https://benchlm.ai/models/llama-4-maverick